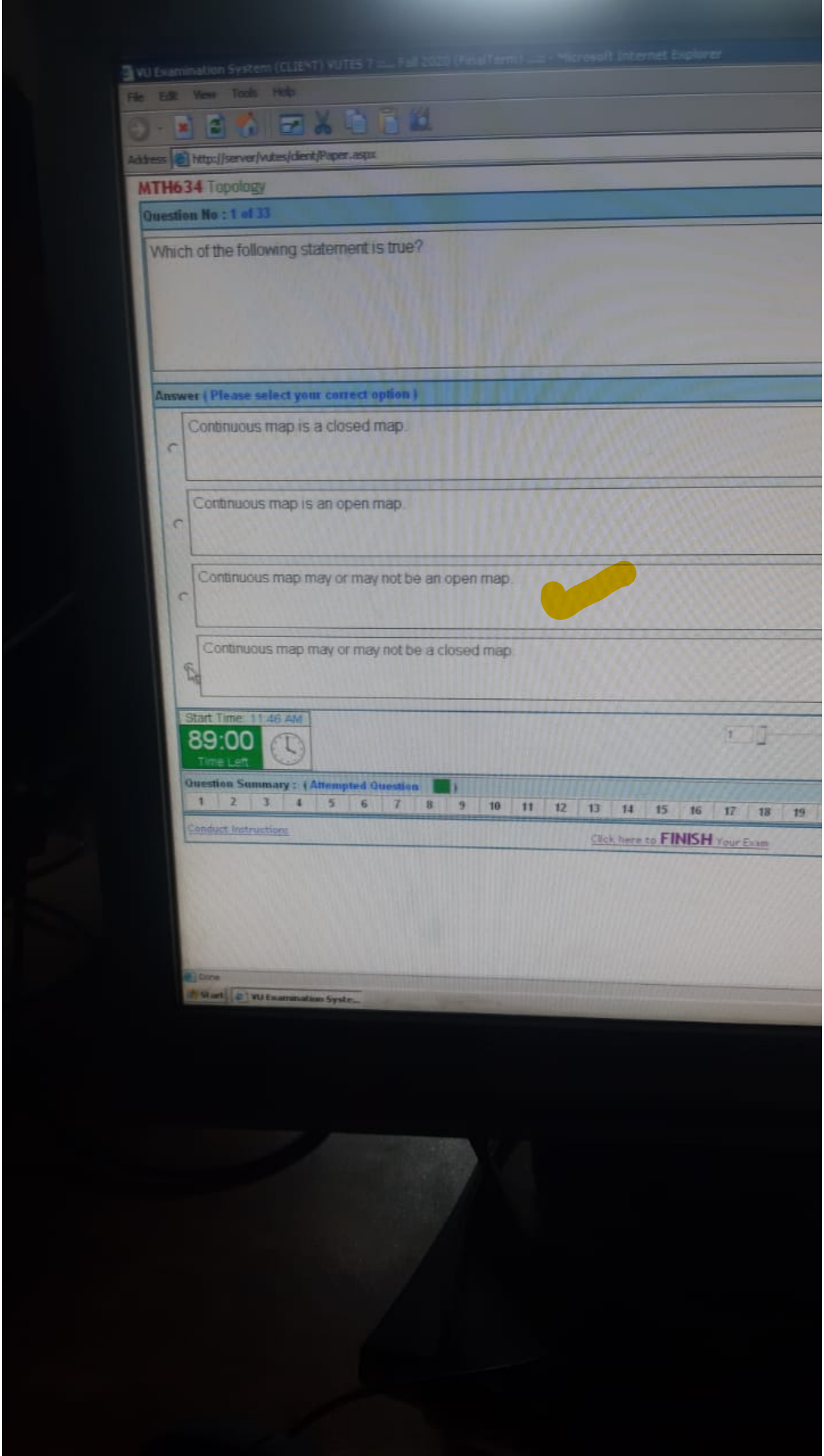

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MTH634 Topology
Question No : 1 of 33
Which of the following statement is true?
Answer (Please select your correct option)
Continuous map is a closed map.
Continuous map is an open map.
Continuous map may or may not be an open map.
Continuous map may or may not be a closed map.
Start Time: 11:46 AM
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Question Summary : (Attempted Question)
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MTH634 Topology

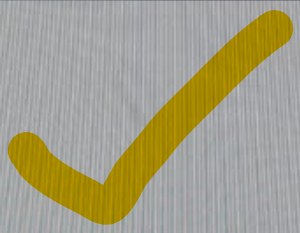
Question No : 22 of 33

A discrete space is not locally connected.

Answer (Please select your correct option)

True

False



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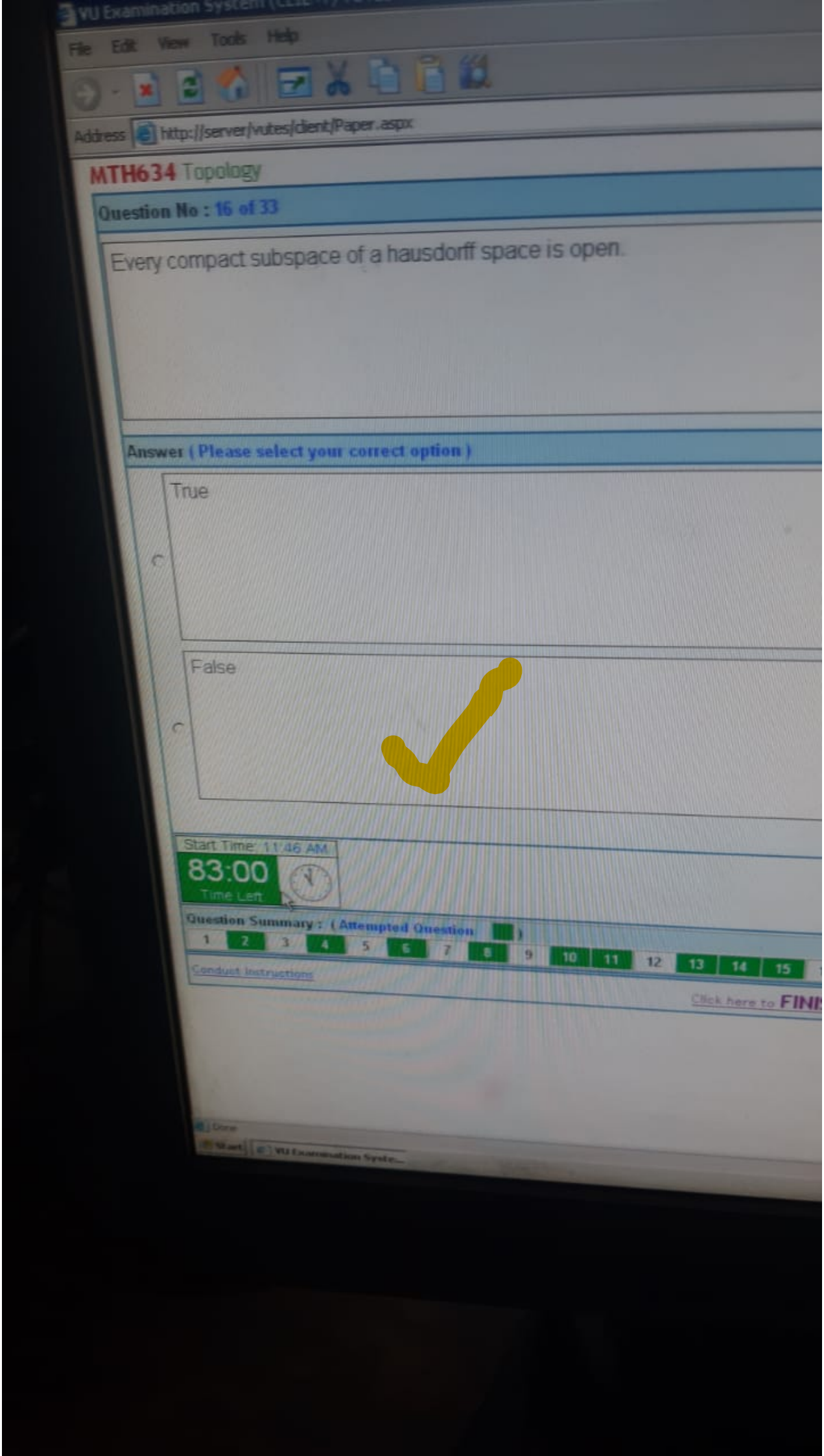
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Question Summary : (Attempted Question)

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Conduct Instructions


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Address http://server/vutes/client/Paper.aspx
MTH634 Topology
Question No : 16 of 33
Every compact subspace of a hausdorff space is open.
Answer (Please select your correct option)
True
False
Start Time: 11:46 AM
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Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Conduct Instructions
Click here to FINI

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MTH634 Topology

Question No : 12 of 33

A regular T_1 – space is called a

Answer (Please select your correct option)

- no isolated points
- T_2 – space
- T_3 – space ✓
- none

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Question Summary : (Attempted Question)

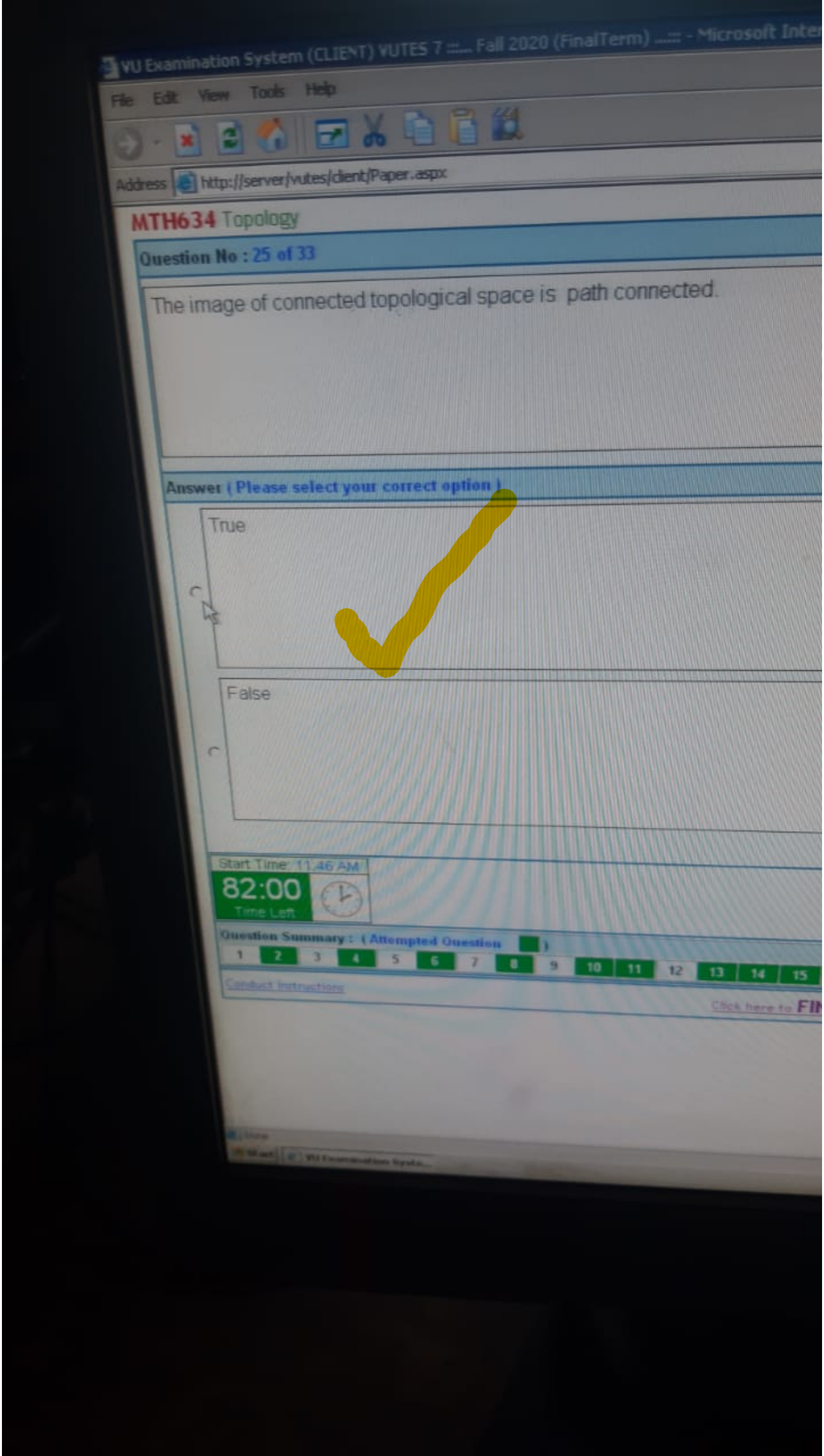
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MTH634 Topology
Question No : 25 of 33
The image of connected topological space is path connected.
Answer (Please select your correct option)
True
False
Start Time: 11:46 AM
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Question Summary : (Attempted Question)
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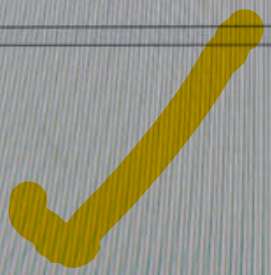
MTH634 Topology

Question No : 17 of 33

If (X,τ) be a compact Hausdorff space then (X,τ) is not normal.

Answer (Please select your correct option)

- True
- False



Start Time 11:46 AM

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Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Conduct Instructions

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MTH634 Topology

Question No : 9 of 33

Which of the following statement is true?

Answer (Please select your correct option)

- Every discrete space is T0 space
- Every discrete space is T_1 space
- Every discrete space is both T_0 and T_1 space
- none

Start Time: 11:46 AM

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Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Conduct Instructions

Click here to F

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MTH634 Topology

Question No : 7 of 33

Let $X = \mathbb{R}$ with usual topology. Which of the following is not an open cover of X?

Answer (Please select your correct option)

- $U = ((-n, n) \mid n \in \mathbb{N})$
- $V = ((n, n+2) \mid n \in \mathbb{Z})$
- $W = ([n, n+1] \mid n \in \mathbb{Z})$
- none

Start Time: 11:46 AM

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Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

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MTH634 Topology

Question No : 3 of 33

Which of the following statement is false?

Answer (Please select your correct option)

- Every metric space is first countable
- Discrete space is first countable
- Every finite set with any topology is first countable
- Cofinite topology on a real line R is first countable

Start Time: 11:46 AM

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Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

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MTH634 Topology

Question No : 5 of 33

For $x, y \in \mathbb{R}$, we define the usual metric on $\mathbb{R}$ as:

Answer (Please select your correct option)

- $d(x,y) = |x - y|$
- $d(x,y) = |\log(y/x)|$
- none

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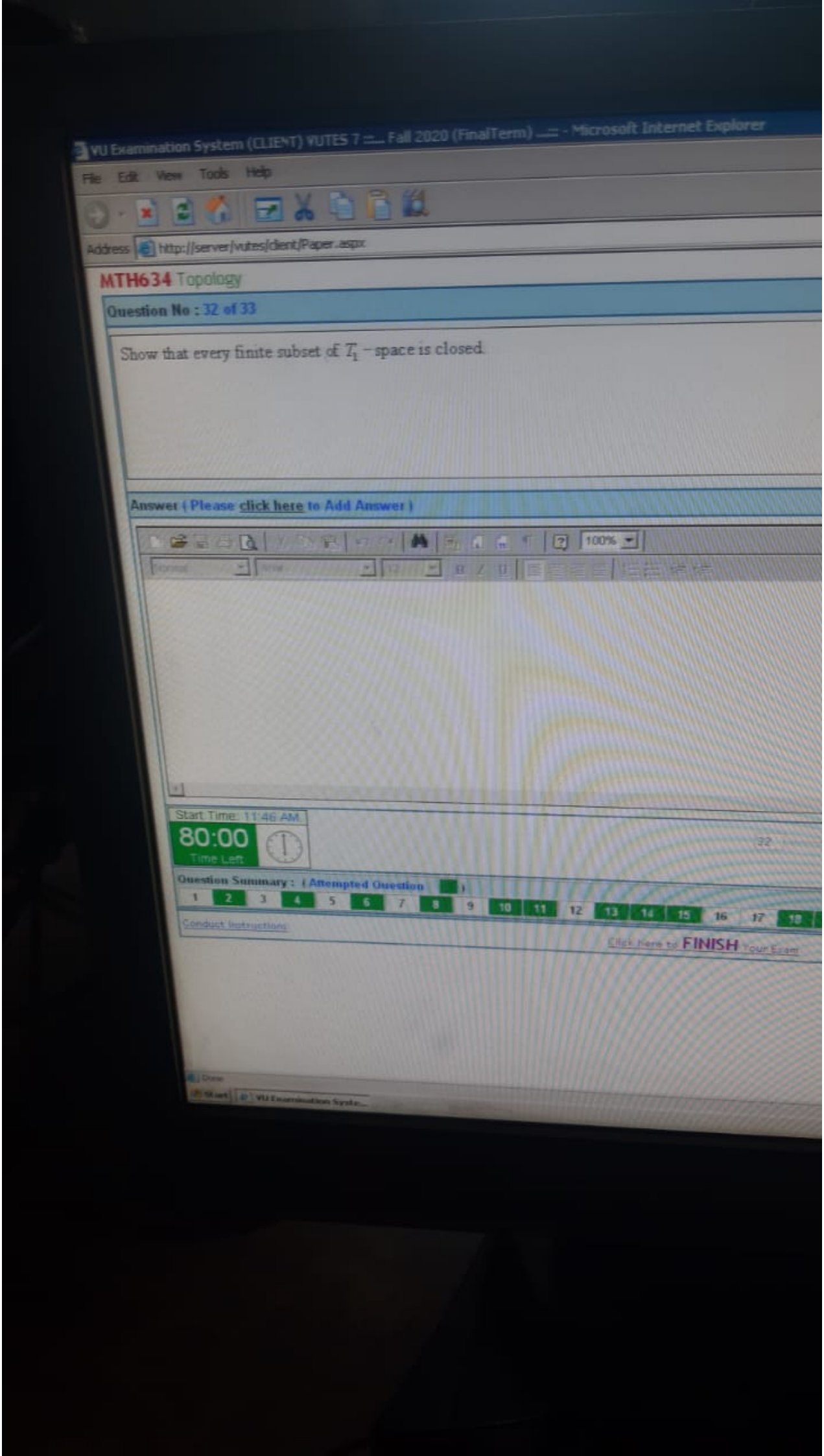
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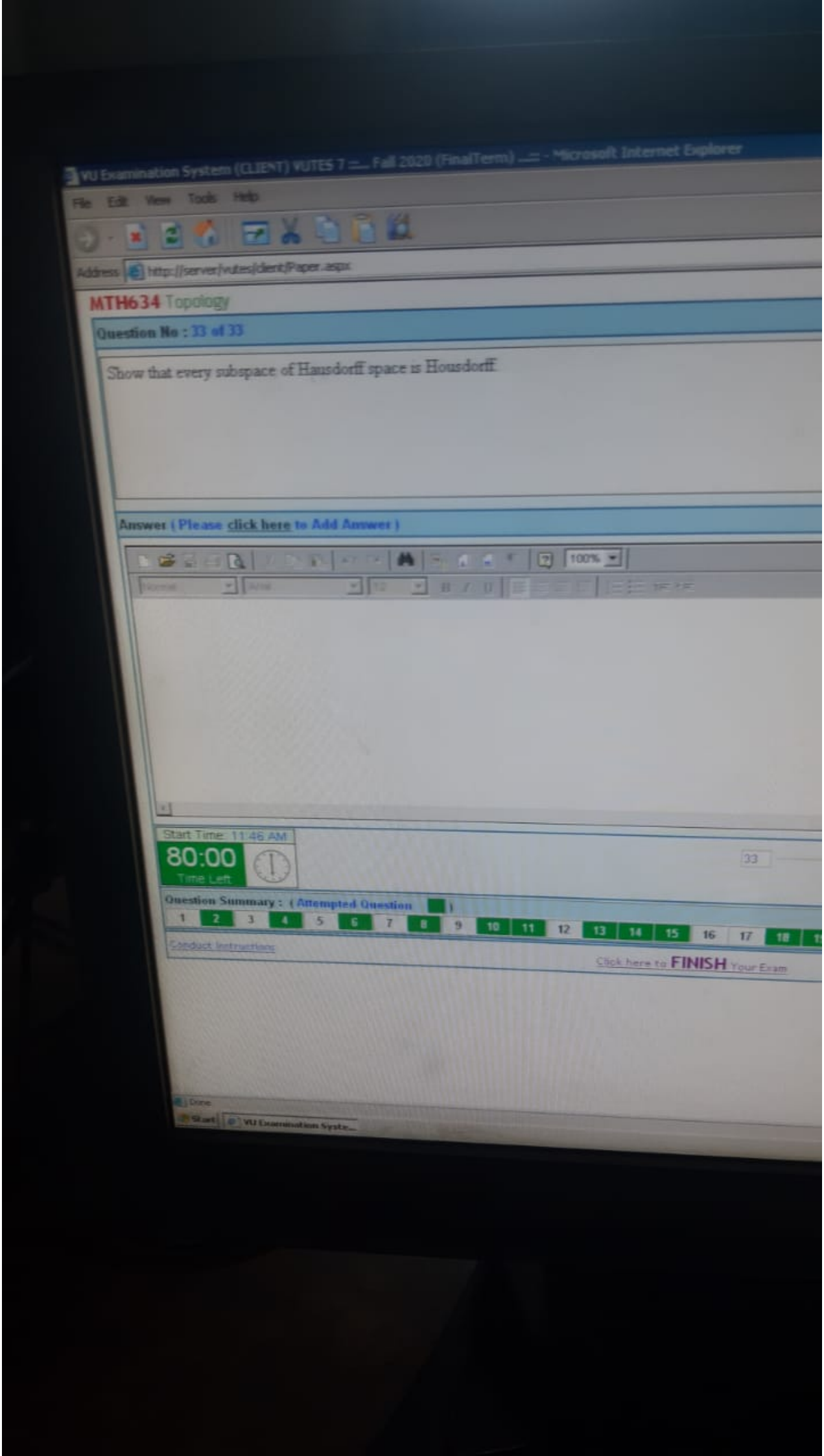
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MTH634 Topology
Question No : 32 of 33
Show that every finite subset of T_1 – space is closed.
Answer (Please click here to Add Answer)
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Question Summary : (Attempted Question)
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MTH634 Topology
Question No : 33 of 33
Show that every subspace of Hausdorff space is Housdorff.
Answer (Please click here to Add Answer)
100%
Start Time: 11:46 AM
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Question Summary : (Attempted Question)
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MTH634 Topology

Question No : 30 of 33

Let τ be a topology on the real line $\mathbb{R}$ generated by the open-closed intervals $(a,b]$. Show that $(\mathbb{R},\tau)$ is Hausdorff.

Answer (Please click here to Add Answer)

Start Time 11:46 AM

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Question Summary : (Attempted Question)

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MTH634 Topology

Question No : 31 of 33

Consider $\mathbb{R}$ with usual topology. Determine whether the sets $B = (0,1)$ and $C = [2,3)$ are separated sets or not ?

Answer (Please click here to Add Answer)

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Start Time: 11:46 AM

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Question Summary : (Attempted Question)

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MTH634 Topology

Question No : 28 of 33

Define locally connected spaces.

Answer (Please click here to Add Answer)

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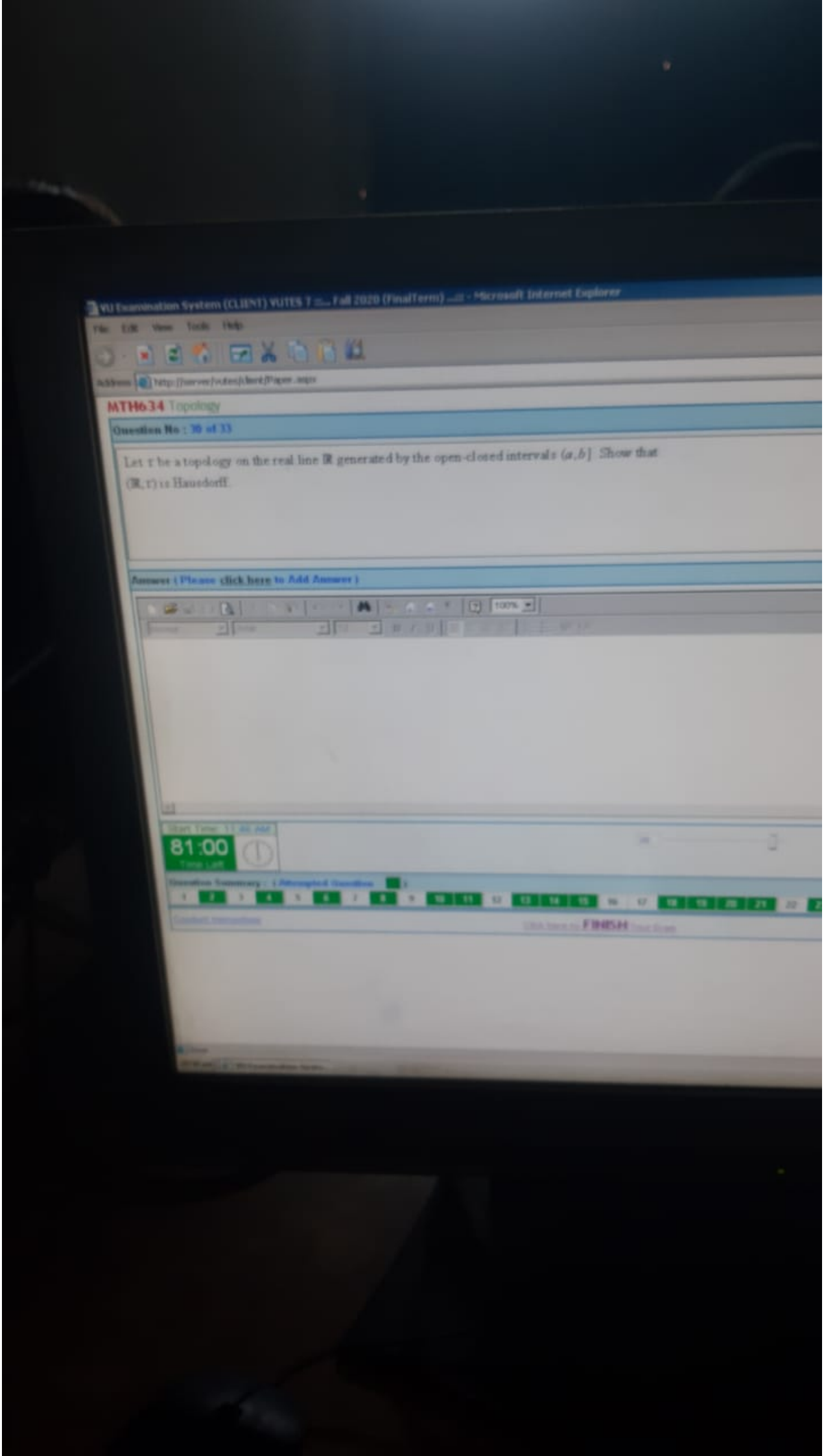
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MTH634 Topology
Question No : 30 of 33
Let τ be a topology on the real line ℝ generated by the open-closed intervals (a,b] . Show that (ℝ,τ) is Hausdorff.
Answer (Please click here to Add Answer)
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MTH634 Topology

Question No : 27 of 33

Define T_1 – space.

Answer (Please click here to Add Answer)

Start Time 11:46 AM

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Question Summary : (Attempted Question)

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MTH634 Topology

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Question No : 29 of 33

Let $X = \{a, b, c, d\}$ with topology $\tau_X = \{\varphi, X\}$ and $Y = \{1,2,3,4\}$ with topology $\tau_Y = \{\varphi, \{1\}, Y\}$. Define a map $f: (X, \tau_X) \to (Y, \tau_Y)$ such that $f(a) = 1, f(b) = 1, f(c) = 1, f(d) = 1$. Show that f is an open and continuous

Answer (Please click here to Add Answer)

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Question Summary : (Attempted Question)

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MTH634 Topology

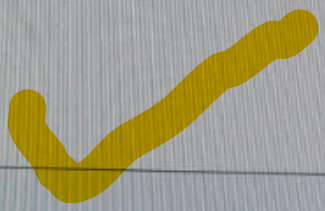
Question No : 22 of 33

A discrete space is not locally connected.

Answer (Please select your correct option)

True

False



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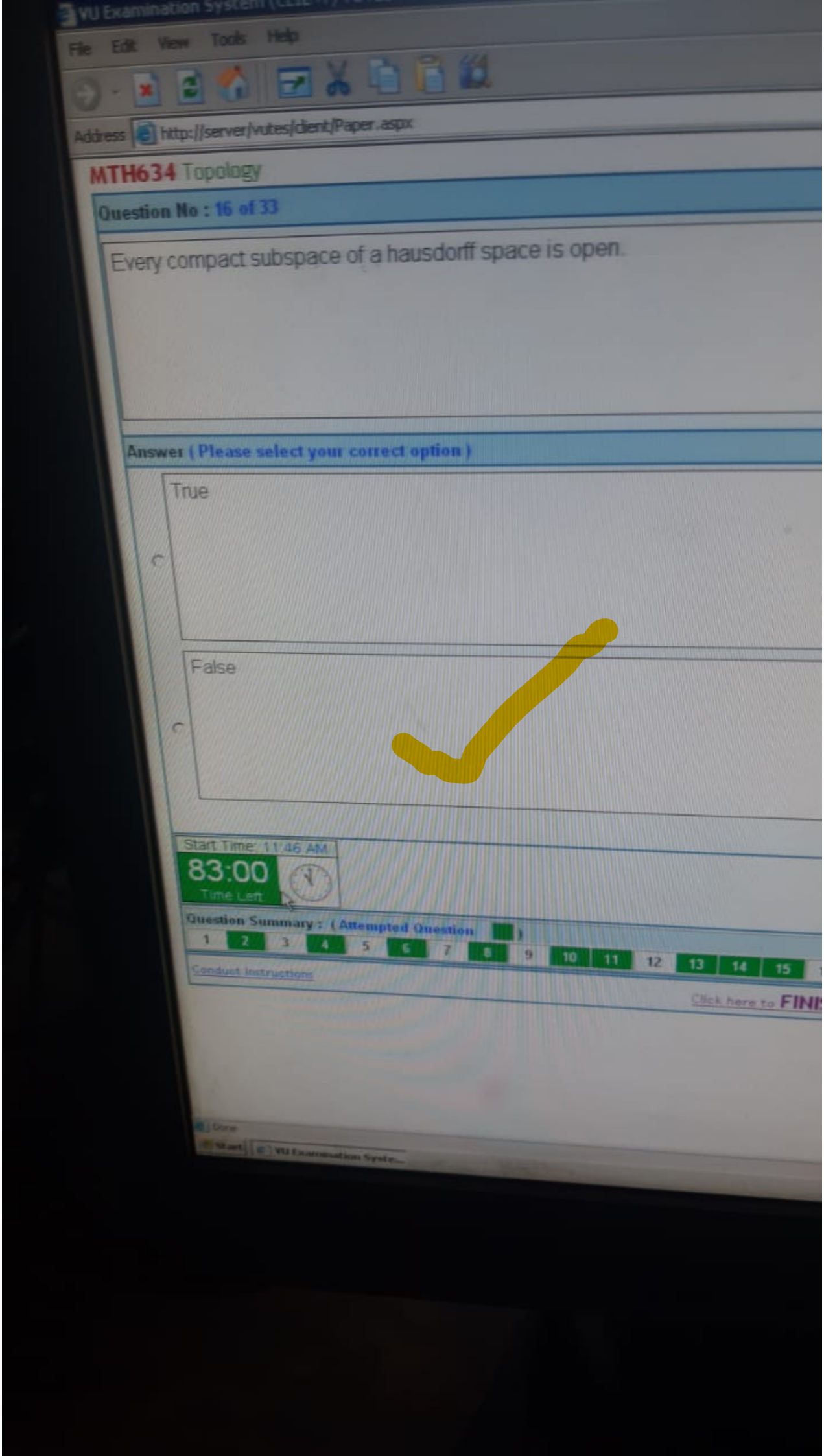
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Question Summary : (Attempted Question)

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MTH634 Topology
Question No : 16 of 33
Every compact subspace of a hausdorff space is open.
Answer (Please select your correct option)
True
False
Start Time: 11:46 AM
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Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
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MTH634 Topology

Question No : 25 of 33

The image of connected topological space is path connected.

Answer (Please select your correct option)

True

False

Start Time: 11:46 AM

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Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

Conduct Instructions

Click here to FIN

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MTH634 Topology

Question No : 12 of 33

A regular T_1 – space is called a

Answer (Please select your correct option)

- no isolated points
- T_2 – space
- T_3 – space
- none

Start Time 11:46 AM

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Question Summary : (Attempted Question)

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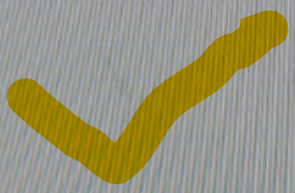
MTH634 Topology

Question No : 17 of 33

If (X,τ) be a compact Hausdorff space then (X,τ) is not normal.

Answer (Please select your correct option)

- True
- False



Start Time 11:46 AM

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Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Conduct Instructions

Click here to FINISH

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MTH634 Topology

Question No : 9 of 33

Which of the following statement is true?

Answer (Please select your correct option)

- Every discrete space is T0 space
- Every discrete space is T_1 space
- Every discrete space is both T_0 and T_1 space
- none

Start Time: 11:46 AM

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Question Summary : (Attempted Question)

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MTH634 Topology

Question No : 5 of 33

For $x, y \in \mathbb{R}$, we define the usual metric on $\mathbb{R}$ as:

Answer (Please select your correct option)

- $d(x,y) = |x - y|$
- $d(x,y) = |\log(y/x)|$
- none

Start Time: 11:46 AM

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Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

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MTH634 Topology

Question No : 7 of 33

Let $X = \mathbb{R}$ with usual topology. Which of the following is not an open cover of X?

Answer (Please select your correct option)

- $U = ((-n, n) \mid n \in \mathbb{N})$
- $V = ((n, n+2) \mid n \in \mathbb{Z})$
- $W = ([n, n+1] \mid n \in \mathbb{Z})$
- none

Start Time: 11:46 AM

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Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Conduct Instructions

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MTH634 Topology

Question No : 3 of 33

Which of the following statement is false?

Answer (Please select your correct option)

- Every metric space is first countable
- Discrete space is first countable
- Every finite set with any topology is first countable
- Cofinite topology on a real line R is first countable

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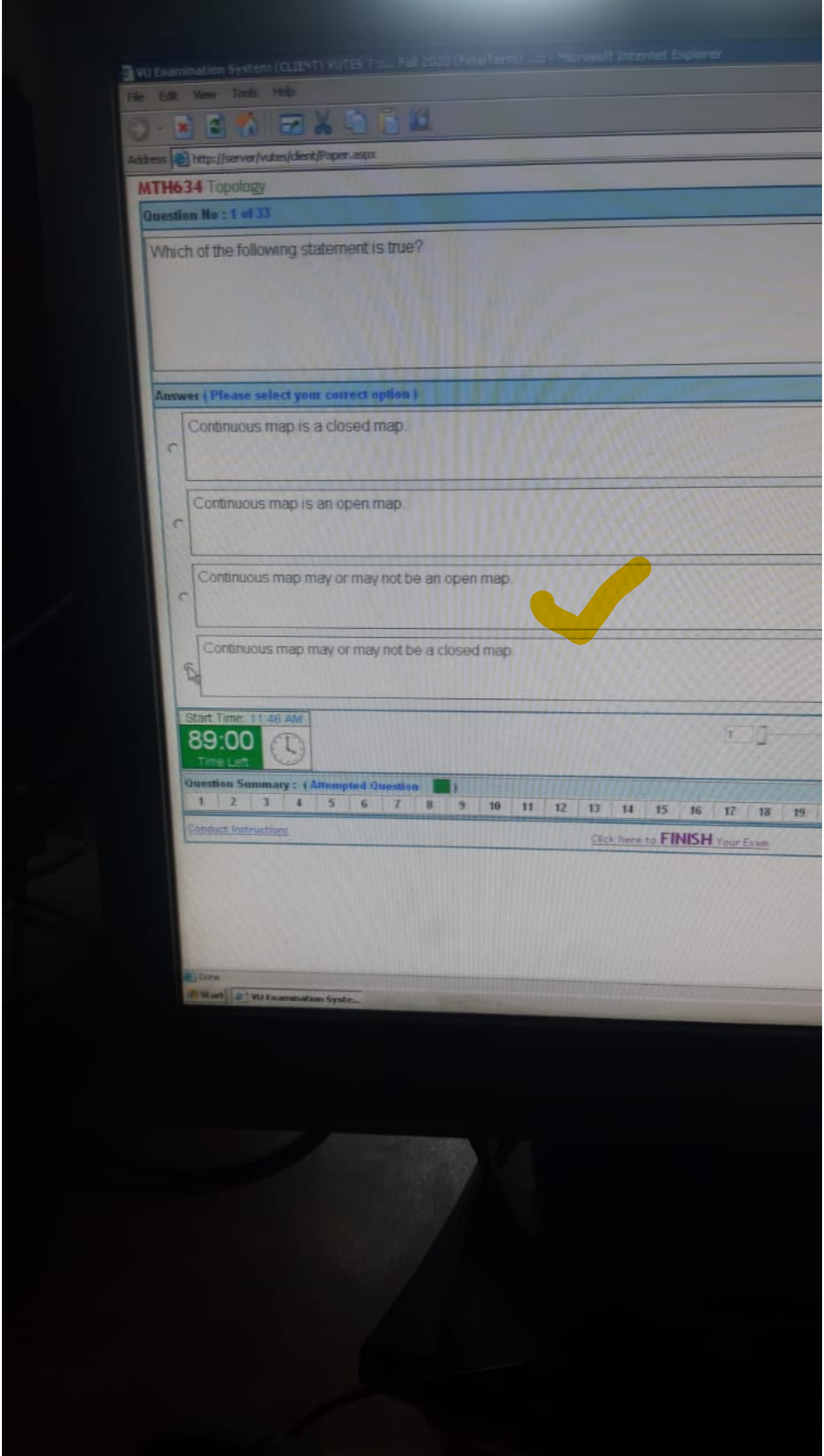
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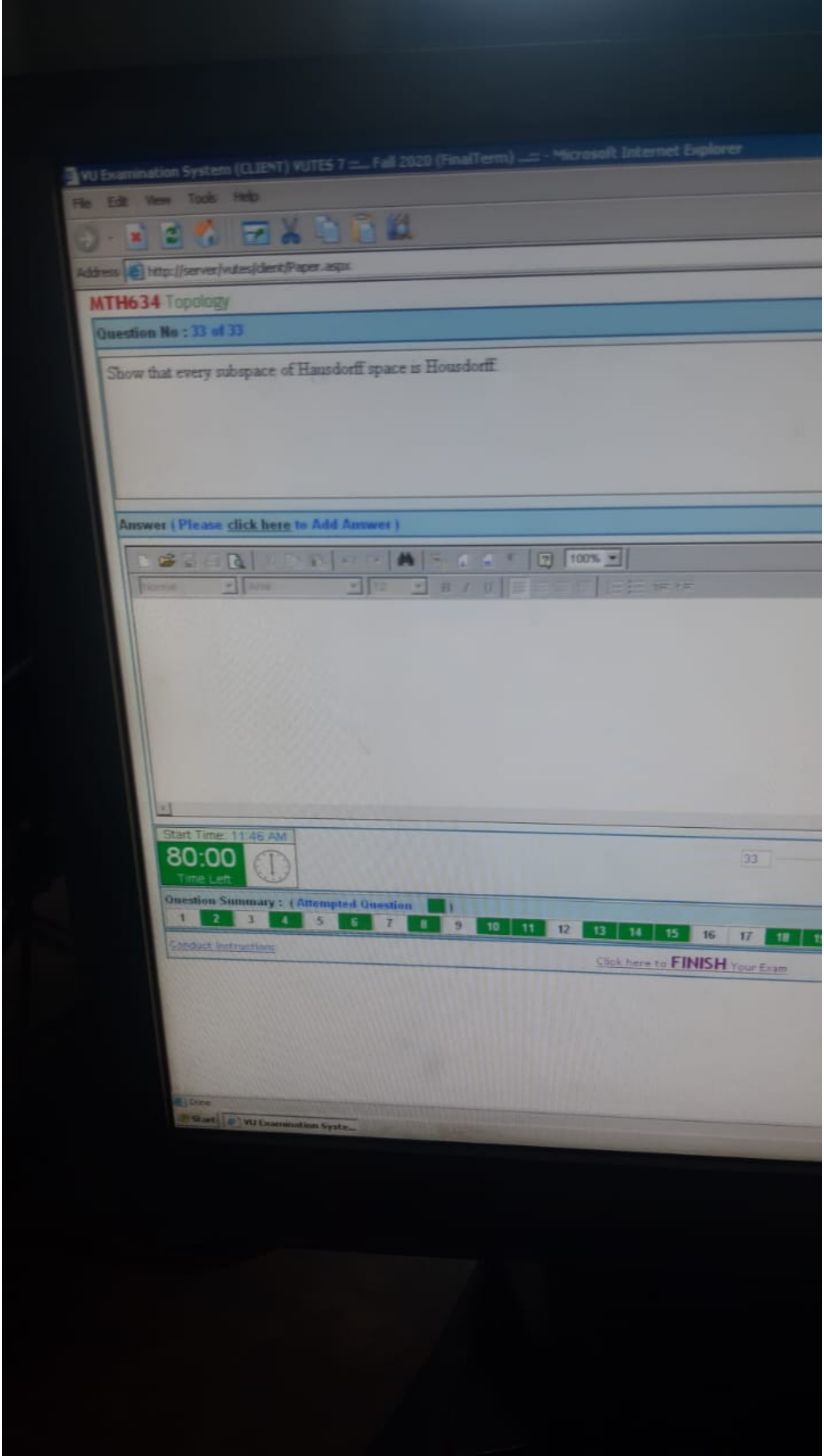
Question Summary : (Attempted Question)

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MTH634 Topology
Question No : 1 of 33
Which of the following statement is true?
Answer (Please select your correct option)
Continuous map is a closed map.
Continuous map is an open map.
Continuous map may or may not be an open map.
Continuous map may or may not be a closed map.
Start Time: 11:46 AM
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Question Summary : (Attempted Question)
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MTH634 Topology
Question No : 33 of 33
Show that every subspace of Hausdorff space is Housdorff.
Answer (Please click here to Add Answer)
100%
Start Time: 11:46 AM
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Question Summary : (Attempted Question)
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MTH634 Topology

Question No : 30 of 33

Let τ be a topology on the real line $\mathbb{R}$ generated by the open-closed intervals $(a,b]$. Show that $(\mathbb{R},\tau)$ is Hausdorff.

Answer (Please click here to Add Answer)

Start Time 11:46 AM

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Time Left

Question Summary : (Attempted Question)

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MTH634 Topology

Question No : 31 of 33

Consider $\mathbb{R}$ with usual topology. Determine whether the sets $B = (0,1)$ and $C = [2,3)$ are separated sets or not ?

Answer (Please click here to Add Answer)

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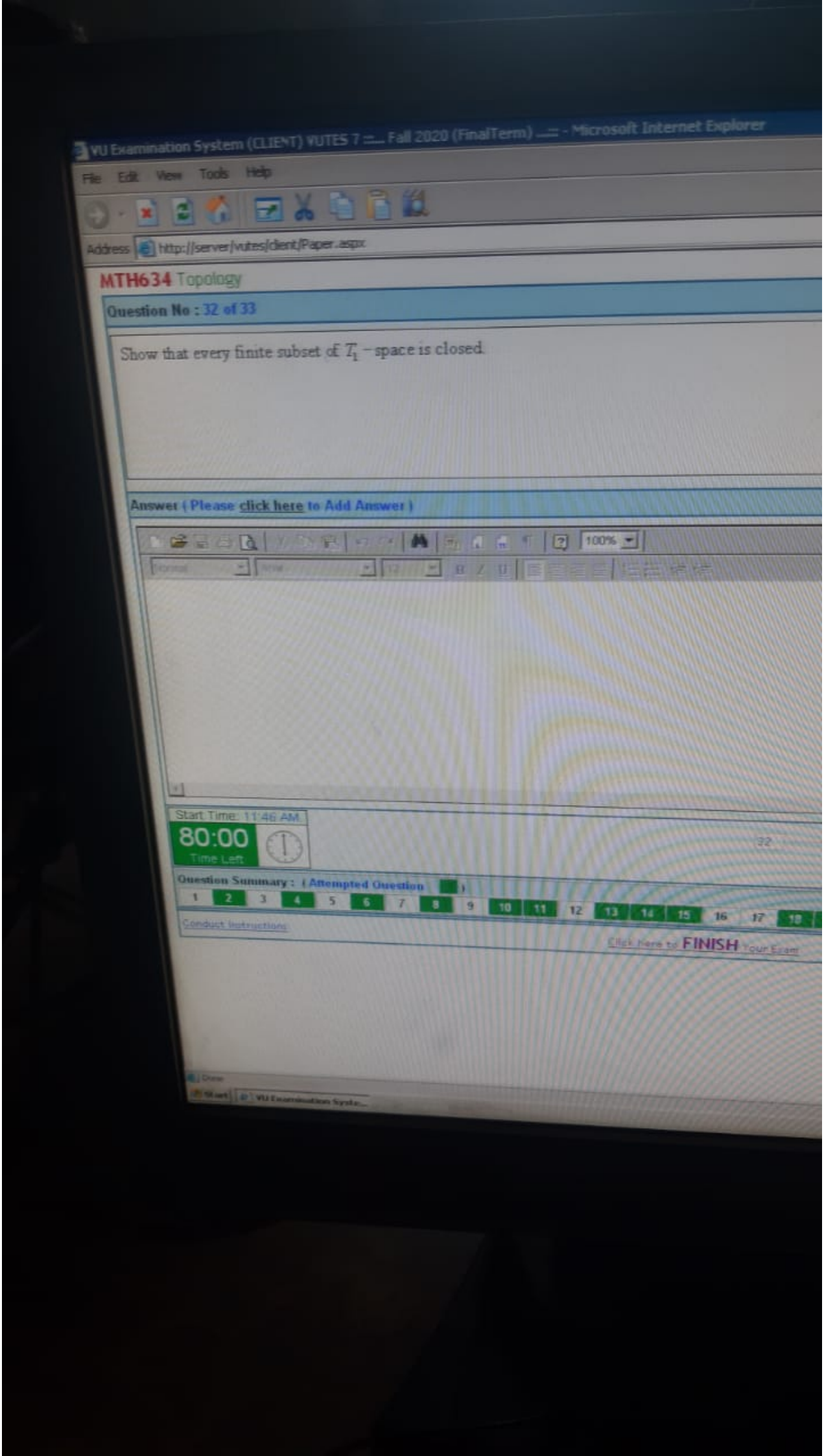
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Question Summary : (Attempted Question)

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MTH634 Topology
Question No : 32 of 33
Show that every finite subset of T_1 – space is closed.
Answer (Please click here to Add Answer)
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Start Time: 11:46 AM
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Question Summary : (Attempted Question)
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VU Examination System (CLIENT) VUTES 7 ..... Fall 2020 (Fin

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Address http://server/vutes/client/Paper.aspx

MTH634 Topology

Question No : 28 of 33

Define locally connected spaces.

Answer (Please click here to Add Answer)

Start Time 11:46 AM

81:00

Time Left

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10

Conduct Instructions

VU Examination System (CLIENT) VUTES 7 — Fall 2020 (FinalTerm) — Microsoft Internet Explorer

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MTH634 Topology

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Question No : 29 of 33

Let $X = \{a, b, c, d\}$ with topology $\tau_X = \{\varphi, X\}$ and $Y = \{1,2,3,4\}$ with topology $\tau_Y = \{\varphi, \{1\}, Y\}$. Define a map $f: (X, \tau_X) \to (Y, \tau_Y)$ such that $f(a) = 1, f(b) = 1, f(c) = 1, f(d) = 1$. Show that f is an open and continuous

Answer (Please click here to Add Answer)

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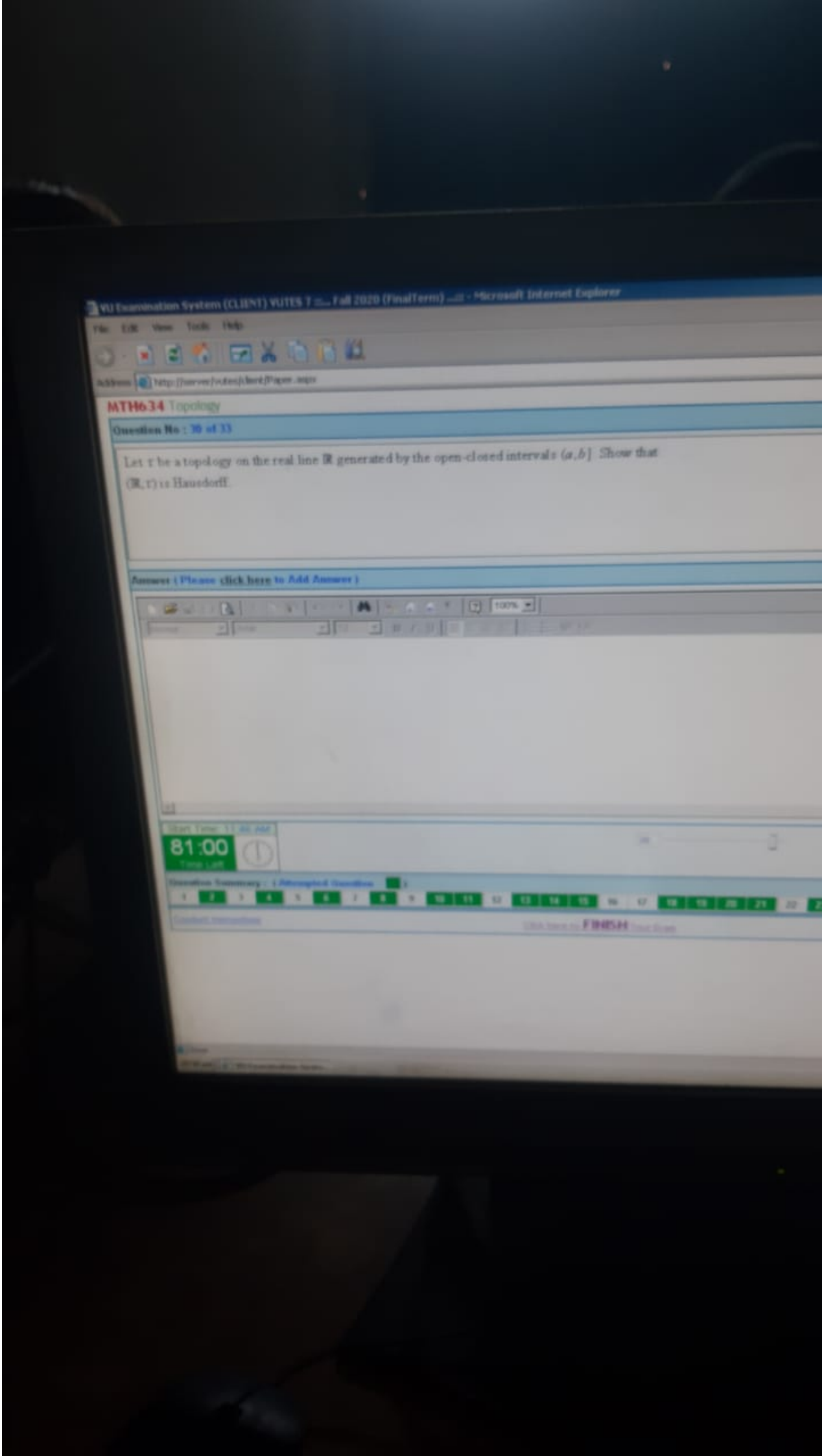
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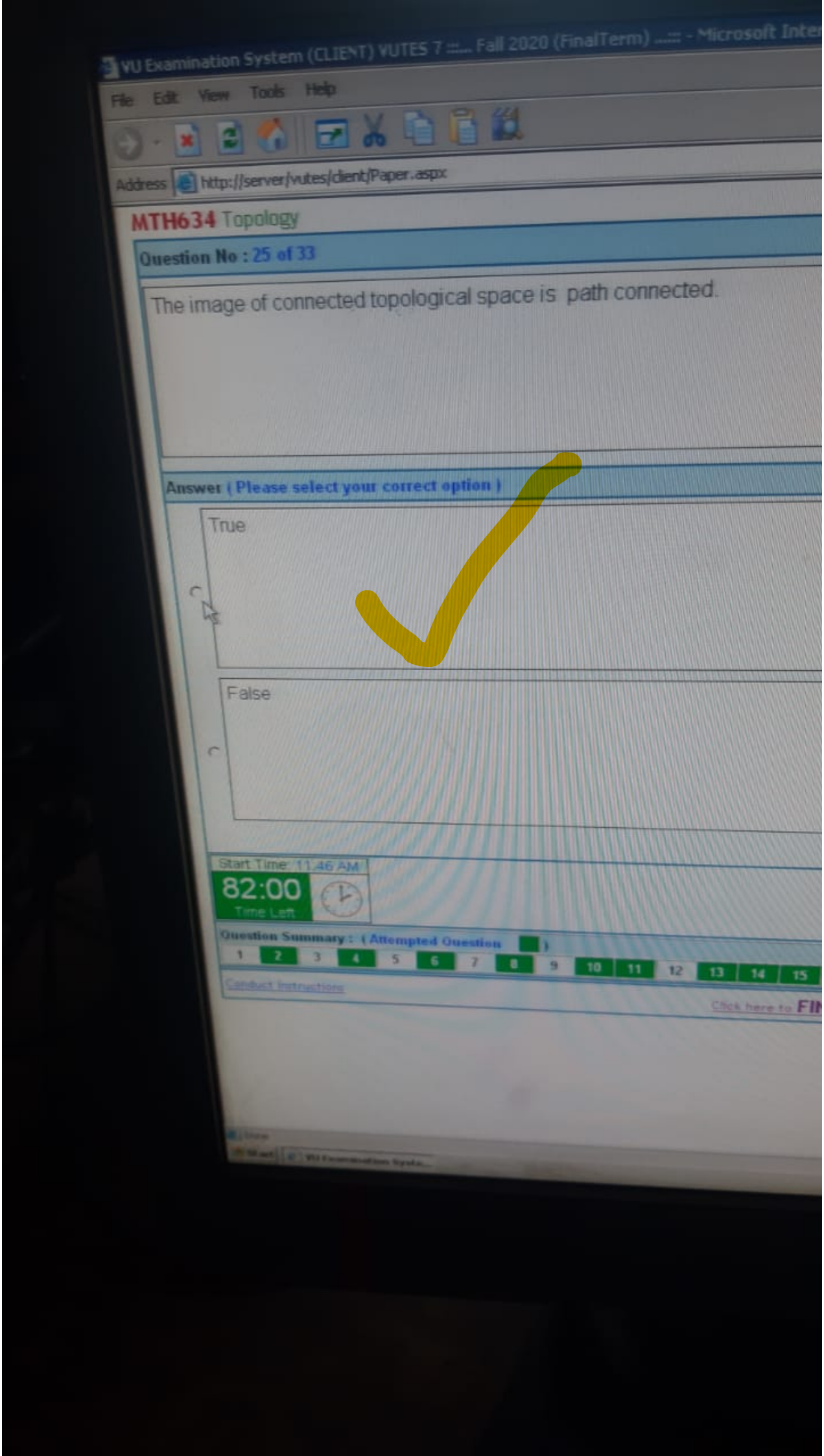
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MTH634 Topology
Question No : 30 of 33
Let τ be a topology on the real line ℝ generated by the open-closed intervals (a,b] . Show that (ℝ,τ) is Hausdorff.
Answer (Please click here to Add Answer)
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VU Examination System (CLIENT) VUTES 7 ..... Fall 2020 (FinalTerm) ..... - Microsoft Inter
File Edit View Tools Help
Address http://server/vutes/client/Paper.aspx
MTH634 Topology
Question No : 25 of 33
The image of connected topological space is path connected.
Answer (Please select your correct option)
True
False
Start Time: 11:46 AM
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Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Conduct Instructions
Click here to FIN

Address http://server/vutes/client/Paper.aspx

MTH634 Topology

Question No : 22 of 33

A discrete space is not locally connected.

Answer (Please select your correct option)

True

False



Start Time: 11:46 AM

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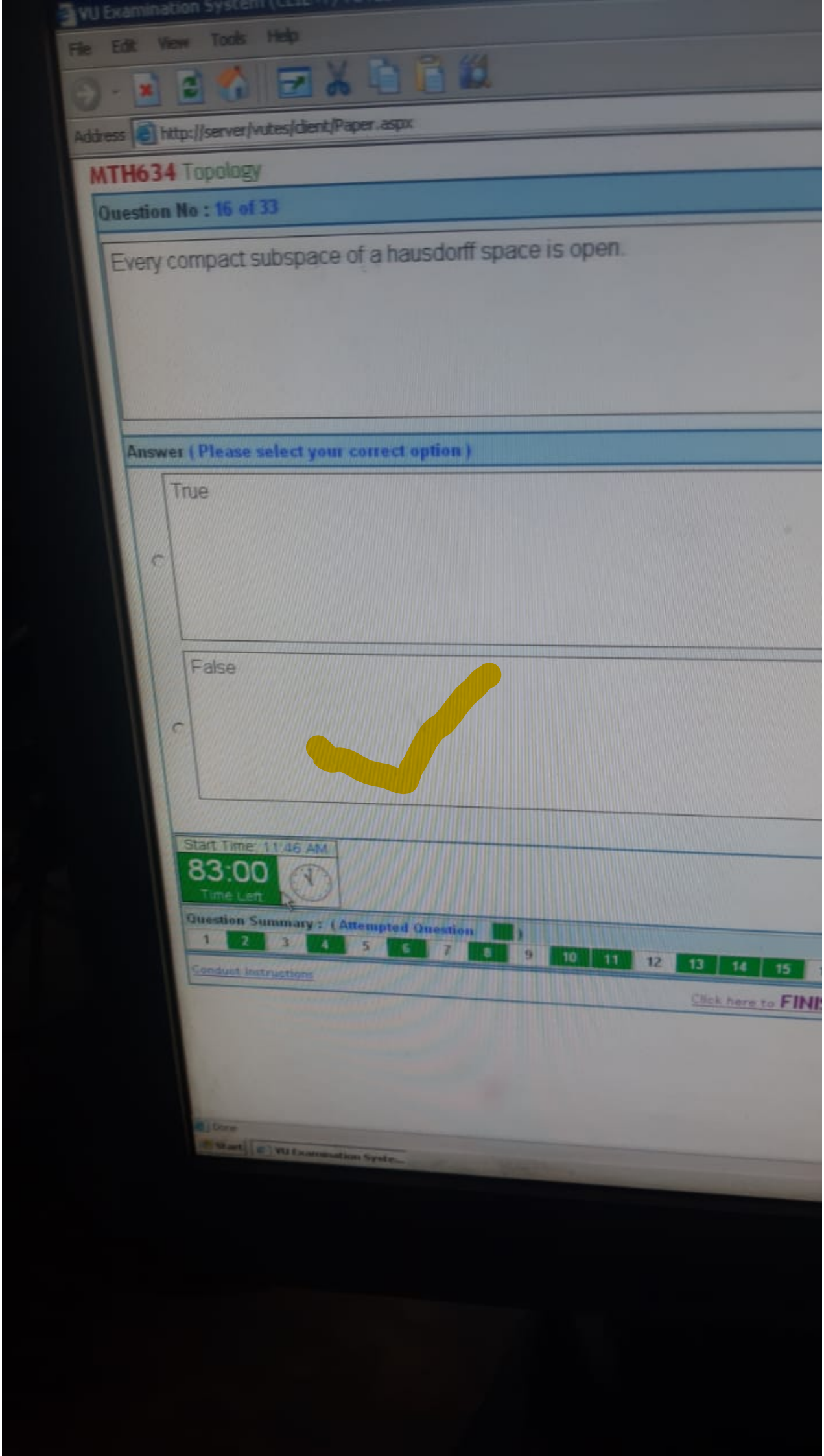
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Click here


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File Edit View Tools Help
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MTH634 Topology
Question No : 16 of 33
Every compact subspace of a hausdorff space is open.
Answer (Please select your correct option)
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False
Start Time: 11:46 AM
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Question Summary : (Attempted Question)
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MTH634 Topology

Question No : 12 of 33

A regular T_1 – space is called a

Answer (Please select your correct option)

- no isolated points
- T_2 – space
- T_3 – space
- none

Start Time 11:46 AM

84:00 Time Left

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14

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MTH634 Topology

Question No : 9 of 33

Which of the following statement is true?

Answer (Please select your correct option)

- Every discrete space is T0 space
- Every discrete space is T_1 space
- Every discrete space is both T_0 and T_1 space
- none

Start Time: 11:46 AM

84:00

Time Left

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

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MTH634 Topology

Question No : 7 of 33

Let $X = \mathbb{R}$ with usual topology. Which of the following is not an open cover of X?

Answer (Please select your correct option)

- $U = ((-n, n) \mid n \in \mathbb{N})$
- $V = ((n, n+2) \mid n \in \mathbb{Z})$
- $W = ([n, n+1] \mid n \in \mathbb{Z})$
- none

Start Time: 11:46 AM

84:00 Time Left

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

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MTH634 Topology

Question No : 5 of 33

For $x, y \in \mathbb{R}$, we define the usual metric on $\mathbb{R}$ as:

Answer (Please select your correct option)

- $d(x,y) = |x - y|$ ✔
- $d(x,y) = |\log(y/x)|$
- none

Start Time: 11:46 AM

84:00 Time Left

Question Summary : (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

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MTH634 Topology

Question No : 3 of 33

Which of the following statement is false?

Answer (Please select your correct option)

- Every metric space is first countable
- Discrete space is first countable
- Every finite set with any topology is first countable
- Cofinite topology on a real line R is first countable

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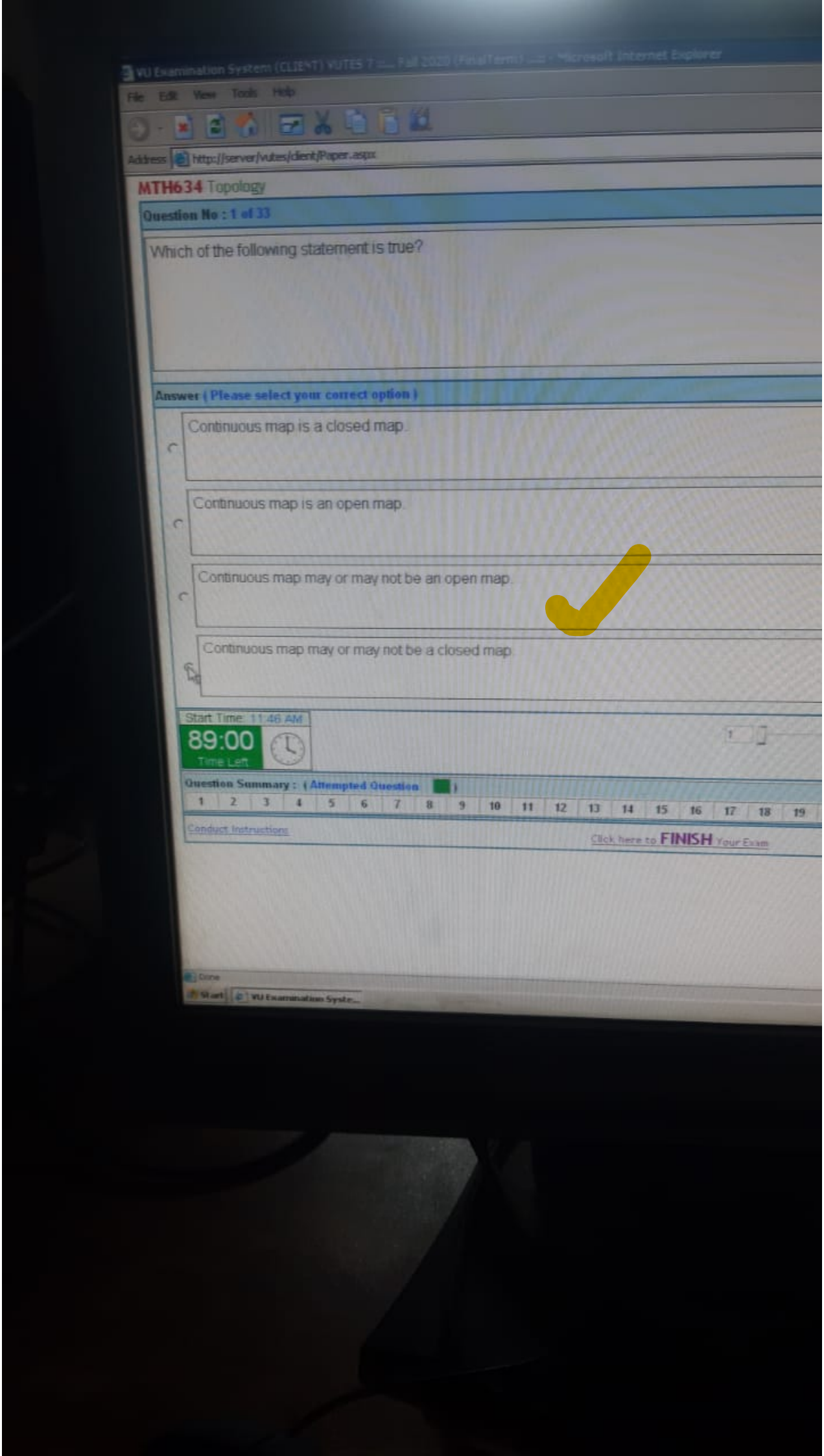
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Question Summary : (Attempted Question)

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Address http://server/vutes/client/Paper.aspx
MTH634 Topology
Question No : 1 of 33
Which of the following statement is true?
Answer (Please select your correct option)
Continuous map is a closed map.
Continuous map is an open map.
Continuous map may or may not be an open map.
Continuous map may or may not be a closed map.
Start Time: 11:46 AM
89:00
Time Left
Question Summary : (Attempted Question)
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19
Conduct Instructions
Click here to FINISH Your Exam
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Start
VU Examination Syste...

Address http://server/vutes/client/Paper.aspx

MTH634 Topology

Question No : 3 of 33

Which of the following statement is false?

Answer (Please select your correct option)

- The set $\mathbb{R}$ with usual topology is metrizable space.
- The set $\mathbb{R}^2$ with usual topology is metrizable space.
- Indiscrete topology on a set X containing more than one point is metrizable

Start Time 9:42 AM

46:00 Time Left

Done

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Topology

Question No : 8 of 33

Which of the following statement is true?

Answer (Please select your correct option)

- Every separable space is second countable.
- Every second countable space is separable.
- $\mathbb{R}$ with lower limit topology is second countable.

Start Time: 9:42 AM

45:00 Time Left

Done

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MTH634 Topology

Question No : 10 of 33

A topological space is a T_1 – space if and only if its each finite subset is

Answer (Please select your correct option)

- a perfect set
- a closed set
- an open set
- none

Start Time: 9:42 AM

45:00 Time Left

10

Done

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Question No : 11 of 33

A finite set has _____ in a T_1 space.

Answer (Please select your correct option)

- no isolated points
- no limit points
- infinte limit points
- none

Start Time: 9:42 AM

45:00
Time Left

Done

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MTH634 Topology

Question No : 2 of 33

Which of the following statement is true?

Answer (Please select your correct option)

- If a map $f : X \to Y$ is bicontinuos then it is also bijective.
- If a map $f : X \to Y$ is bicontinuos then it is not bijective.
- If a map $f : X \to Y$ is bicontinuos then it may or may not be bijective.

Start Time: 9:42 AM

46:00 Time Left

2

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MTH634 Topology

Question No : 9 of 33

Which of the following statement is true?

Answer (Please select your correct option)

- A separable metric space is second countable. ✓
- A separable metric space is first countable.
- Every metric space is second countable.
- none

Start Time: 9:42 AM

45:00
Time Left

9

Done

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Question No : 6 of 33 Marks: 1 (Bu

Let $B_1(x,r_1)$ and $B_2(x,r_2)$ be two open balls centered at x with radii r_1 and r_2 respectively. Then the following statement is true.

Answer (Please select your correct option)

- $B_1(x,r_1) = B_2(x,r_2)$
- either $B_1(x,r_1) \subset B_2(x,r_2)$ or $B_1(x,r_1) \supset B_2(x,r_2)$ ✓
- none

Start Time: 9:42 AM

46:00 Time Left

Done

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Question No : 5 of 33

For metric d on X, an open ball of radius r with center $x \in X$ is defined as:

Answer (Please select your correct option)

- $B(x,r) = \{y \in X \mid d(x,y) \leq r\}$
- $B(x,r) = \{y \in X \mid d(x,y) > r\}$
- $B(x,r) = \{y \in X \mid d(x,y) < r\}$ ✓
- none

Start Time: 9:42 AM

46:00 Time Left

Done

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MTH634 Topology

Question No : 4 of 33

Which of the following statement is false?

Answer (Please select your correct option)

- Discrete topology on a countable set X is second countable
- Any finite set with any topology is second countable
- Discrete topology on a real line R is second countable
- The set R with usual topology is second countable

Start Time: 9:42 AM

46:00 Time Left

Done

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MTH634 Topology

Question No : 1 of 33

Which of the following statement is true?

Answer (Please select your correct option)

- Projection map may or may not be closed.
- Projection map is open.
- Projection map is closed.

Start Time: 9:42 AM

47:00 Time Left

Done

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MTH634 Topology

Question No : 24 of 33 **M**

If (X,τ) be a path connected topological space, then (X,τ) is connected

Answer (Please select your correct option)

- True
- False

Start Time 9:42 AM

49:00

Time Left

24

Done

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MTH634 Topology

Question No : 25 of 33

The image of connected topological space is path connected.

Answer (Please select your correct option)

- True
- False

Start Time: 9:42 AM

49:00

Time Left

25

Done

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MTH634 Topology

Question No : 32 of 33 Mark

Show that any subspace (Y, τ_Y) of a first countable space (X, τ) is also first countable.

Answer (Please click here to Add Answer)

100%

Start Time: 9:42 AM

47:00

Time Left

32

Done

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Question No : 12 of 33

A regular T_1 – space is called a

Answer (Please select your correct option)

- no isolated points
- T_2 – space
- T_3 – space
- none

Start Time: 3:42 AM

75:00
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Question Summary : 1 Attempted Question

Done

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Question No : 11 of 33

A finite set has _____ in a T_1 space

Answer (Please select your correct option)

- no isolated points
- no limit points
- infinte limit points
- none

Start Time 9:42 AM

75:00
Time Left

Question Summary : (Attempted Question)

Done

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Question No : 4 of 33

Which of the following statement is false?

Answer (Please select your correct option)

- Discrete topology on a countable set X is second countable.
- Any finite set with any topology is second countable
- Discrete topology on a real line R is second countable
- The set $\mathbb{R}$ with usual topology is second countable

Start Time 9:42 AM

76:00
Time Left

Question Summary : (Attempted Question)

Done

Start VU Examination Syste...

Address http://server/vutes/clientPaper.aspx

Question No : 5 of 33

For metric d on X, an open ball of radius r with center $x \in X$ is defined as:

Answer (Please select your correct option)

- $B(x,r) = \{y \in X \mid d(x,y) \leq r\}$
- $B(x,r) = \{y \in X \mid d(x,y) > r\}$
- $B(x,r) = \{y \in X \mid d(x,y) < r\}$
- none

Start Time: 9:42 AM

76:00
Time Left

Question Summary : (Attempted Question)

Done

Start | VU Examination Syste...

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Question No : 6 of 33 **Marks: 1 (Budg**

Let $B_1(x,r_1)$ and $B_2(x,r_2)$ be two open balls centered at x with radii r_1 and r_2 respectively. Then the following statement is true

Answer : Please select your correct option :

- $B_1(x,r_1) = B_2(x,r_2)$
- either $B_1(x,r_1) \subset B_2(x,r_2)$ or $B_1(x,r_1) \supset B_2(x,r_2)$
- none

76:00
Time Left

Question Summary : (Attempted (Correct))

Question No : 8 of 33

Which of the following statement is true?

Answer (Please select your correct option)

- Every separable space is second countable
- Every second countable space is separable
- R with lower limit topology is second countable

Start Time: 9:42 AM

76:00
Time Left

Question Summary : 1 Attempted Question

Done

Start | VU Examination Syste...

Address http://server/vutes/client/Paper.aspx

Question No : 9 of 33

Which of the following statement is true?

Answer (Please select your correct option)

- A separable metric space is second countable.
- A separable metric space is first countable.
- Every metric space is second countable
- none

Start Time 9:42 AM

76:00
Time Left

Question Summary : (Attempted Question)

Done

Start | VU Examination Syste...

Address http://server/vutes/client/Paper.aspx

Question No : 10 of 33

A topological space is a T_1 – space if and only if its each finite subset is

Answer (Please select your correct option)

- a perfect set
- a closed set
- an open set
- none

Start Time 9:42 AM

75:00 Time Left

Question Summary : 1 Attempted Question

Done

Start | VU Examination Syste...

Address http://server/vutes/client/Paper.aspx

Question No : 3 of 33

Which of the following statement is false?

Answer (Please select your correct option)

- The set $\mathbb{R}$ with usual topology is metrizable space
- The set $\mathbb{R}^2$ with usual topology is metrizable space
- indiscrete topology on a set X containing more than one point is metrizable

Start Time 3:42 AM

77:00 Time Left

Question Summary : 1 Attempted Question

Done

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Address http://server/vutes/client/Paper.aspx

Question No : 1 of 33

Which of the following statement is true?

Answer (Please select your correct option)

- Projection map may or may not be closed
- Projection map is open
- Projection map is closed

Start Time: 9:42 AM

77:00
Time Left

Question Summary : 1 Attempted Question

Done

Start | VU Examination Syste...

Question No : 23 of 30

A real line $\mathbb{R}$ with usual topology is path connected.

Answer (Please select your correct option)

- True
- False

Start Time 8:42 AM

79:00
Time Left

Question Summary : 1 Answered Question

Done

Start | VU Examination Syste...

A discrete space is not locally connected.

Answer (Please select your correct option)

- True
- False

Start Time: 8:42 AM

79:00
Time Left

Question Summary - 1 Attempted Question

Done

Start | VU Examination Syste...

Question No : 25 of 31

The image of connected topological space is path connected

Answer (Please select your correct option)

- True
- False

Start Time 9:42 AM

79:00
Time Left

Question Summary : 1 Answered Question

Done

Start | VU Examination Syste...

Question No : 19 of 33

Closure of a connected subset is also connected.

Answer (Please select your correct option)

True

False



Start Time: 8:42 AM

80:00

Time Left

Address http://server/vutes/clientPaper.aspx

Question No : 20 of 33

Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined as $f(x) = x + 1$. Fixed point of f is:

Answer (Please select your correct option)

- 0
- 1
- 2
- no fixed point

Start Time: 9:42 AM

80:00
Time Left

Question Summary : (Attempted Question)

Done

Start | VU Examination Syste...

Address http://server/vuest/client/Paper.aspx

Question No : 21 of 33

Let C be a connected component of a topological space X, then

Answer (Please select your correct option)

- C is open
- C is closed
- C is infinite
- none

Start Time 9:42 AM

79:00
Time Left

Question Summary : (Attempted Question)

Done

Start | VU Examination Syste...

Address http://server/vutes/client/Paper.aspx

Question No : 24 of 33

If (X,τ) be a path connected topological space, then (X,τ) is connected

Answer (Please select your correct option)

- True
- False

Start Time 9:42 AM

79:00
Time Left

Question Summary - Attempted Question

Done

Start | VU Examination Syste...

Question No : 16 of 33

Every compact subspace of a hausdorff space is open

Answer (Please select your correct option)

- True
- False

Start Time: 9:42 AM

81:00
Time Left

Done

Start

Address http://server/vulms/client/Paper.aspx

Question No : 27 of 33

Define Normal space.

Answer (Please click here to Add Answer)

Start Time 9:42 AM

78:00
Time Left

Question Summary (Attempted Question)

1 2 3 4 5 6 7 8 9 10 11 12

Done

Address http://server/vutes/client/Paper.aspx

Question No : 26 of 33

Define metric topology.

Answer (Please click here to Add Answer)

Start Time 9:42 AM

79:00
Time Left

Question Summary : 1 Attempted Question

Done

Start VU Examination Syste...

Address http://server/vutes/client/Paper.aspx

MTH634 Topology

Question No : 30 of 33

Show that a set X with topology T containing finite number subsets of X is compact.

Answer (Please click here to Add Answer)

100%

Start Time: 9:42 AM

48:00

Time Left

30

Done

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Question No : 31 of 33

Define connected set

Answer (Please click here to Add Answer)

100%

Start Time: 9:42 AM

48:00

Time Left

Done

Start

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MTH634 Topology

Question No : 29 of 33

Consider the sets $X = (1,2,3)$ and $Y = (a,b,c)$ with topologies $\tau_X = (\phi,(1),(1,2),X)$ and $\tau_Y = (\phi,(a,b),(b),Y)$ repectively. Define a map $f: X \to Y$ as:

$f(1) = f(2) = a, f(3) = c,$

Show that f is a closed map.

Answer (Please click here to Add Answer)

100%

Start Time: 9:42 AM

48:00

Time Left

29

Done

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MTH634 Topology

Question No : 28 of 33

Define connected component in a topological space X.

Answer (Please click here to Add Answer)

100%

Start Time 9:42 AM

48:00

Time Left

26

Done

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Question No : 27 of 33

Define Normal space.

Answer (Please click here to Edit Answer)

In topology and related branches of mathematics a normal space is a topologi
two disjoint closed sets of X have disjoint open neighborhood. a normal hausd

Start Time 9:42 AM

48:00

Time Left

27

Done

Start

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MTH634 Topology

Question No : 26 of 33

Define metric topology.

Answer (Please click here to Edit Answer)

100%

Metric topology: let x be a non empty set with metric d. The topology t or
with respect to d is called metric topology.

Start Time 9:47 AM

48:00
Time Left

26

Done

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Question No : 33 of 33 Marks: 5 (Budgeted Time Mi

Consider $X = (a,b,c,d,e)$ with topology $\tau = (\phi,(a),(a,b),(a,b,c),(a,d,e),(a,b,d,e),X)$ Show that the set $A = (c,e)$ is disconnected subset of X.

Answer (Please click here to Add Answer)

